Pioneer academics

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Pioneer Research Institute Evaluation Form

Scholar Information

Scholar's Full Name:	Xu, Haorong
Scholar's Research Concentration:	The Practical Challenges of Writing Secure Code
Title of Scholar's Research Paper:	SoK: Recent Developments in Hardware-Assisted Control Flow Integrity

Grading and Academic Oversight Section

Grade conferred by professor: A+

**Full grading rubric approved by Pioneer Academics and Oberlin College*

The Pioneer Research Institute's college accreditation and academic oversight are conducted collaboratively with Oberlin College:

Listing on Common Application:	"Summer Program, Credit Awarded Directly by Oberlin College"
Course Title on Coalition Application:	"099" "Pioneer Research Course"
College Course ID on UC Application:	"INST 099" "Pioneer Research Course"

Comprehensive Evaluation Section

Evaluator Name & Full Title:	Hanan Hibshi, Assistant Teaching Professor
College/University & Department:	Carnegie Mellon University, College of Engineering, The Information Networking Institute
Full Educational Background of Evaluator:	B.S. in Computer Science, King Abdul-Aziz University (2003) M.S in Information Security Technology and Management, Carnegie Mellon University (2011) Ph.D. in Societal Computing, Carnegie Mellon University (2018)

Please briefly explain the nature and requirements of the research paper and your interaction with the scholar:

Scholars are expected to attend four group sessions along with five other scholars in the group. Scholars completed assigned readings before each group session. During group sessions, scholars learned about computer security code vulnerabilities and the challenges of writing secure code. In each session, scholars would prepare to respond to a discussion question for that session.

After that, scholars attended five one-on-one meetings with the instructor. Each meeting had a milestone to be completed by the scholar. The milestones start from writing a research proposal and an outline to showing a final draft of the paper at the last one-on-one meeting.

Finally, scholars meet again in a final group session where each scholar presents their research and responds to questions from other scholars.

Haorong attended all group sessions and was an active participant in discussions.

I worked with scholars individually on their research projects during one-on-one meetings. Haorong selected an interesting and challenging topic by conducting a systematic literature review on Recent Developments in Hardware-Assisted Control Flow Integrity. This topic is typically explored at the upper-level undergraduate or graduate level. Haorong was responsible for all aspects of the project, from selecting the research direction and reviewing the literature to designing experiments and analyzing results. He consistently demonstrated exceptional productivity, making substantial progress at every meeting. Throughout the research process, Haorong exhibited strong technical abilities, intellectual curiosity, and considerable potential as a researcher.

Haorong presented his work during the final group session of the program. He began with an engaging question that encouraged audience participation and immediately captured attention. His presentation was well organized, with a clear flow of ideas and a strong command of voice. He did an excellent job simplifying a complex hardware security topic for a broad audience and demonstrated a deep understanding of the subject through thoughtful responses to questions. Haorong also actively engaged with his peers by asking insightful questions during their presentations, reflecting genuine curiosity and a strong interest in research. Overall, his presentation stood out for its clarity, engagement, and technical depth.

Please rate the scholar in the following areas:

	Excellent	Good	Average	Below Average	Poor
Ability to form original ideas and concepts	X				
Ability to communicate thoughts in an effective and articulate manner	X				
Ability to synthesize and organize information from disparate sources	X				
Level of scholar's curiosity, aptitude, and industriousness	X				

Based on the scholar's performance throughout Pioneer, how would you rate this scholar's potential for undergraduate-level academic work at a top college/university, relative to other undergraduate students whose academic work you have mentored?

One of the best I have encountered, top 5% X	Excellent, top 10%
Very Good, top 25%	Good, top 50%
Below Average	

Describe some of the scholar's strengths:

Haorong is an exceptional student whose academic performance, intellectual maturity, and technical abilities place him among the strongest students I have worked with. Throughout the project, he demonstrated a high degree of independence, taking ownership of the research process and consistently producing high-quality work with minimal guidance. He showed strong analytical thinking, excellent problem-solving skills, and the ability to quickly master complex technical concepts.

What distinguishes Haorong most is his rare combination of technical excellence, intellectual curiosity, independence, and professionalism. Our discussions were consistently thoughtful and insightful, reflecting a level of understanding often seen in advanced undergraduate or graduate students. He approaches challenges with rigor and creativity, incorporates feedback effectively, and maintains exceptionally high standards in his work. I am confident that Haorong has the skills, maturity, and potential to thrive in a highly selective academic environment and make meaningful contributions to research and innovation.

Describe some areas in which the scholar can improve:

A positive area for Haorong's continued growth is developing greater patience when discussing ideas in group settings. Haorong is an exceptionally quick thinker who often understands concepts rapidly and is eager to contribute to the conversation. At times, this enthusiasm can lead him to jump in before others have fully developed or completed their thoughts. This is a common trait among highly capable and intellectually engaged students, particularly those who process information quickly. As Haorong gains more experience in collaborative academic and research environments, he will continue to refine his ability to balance active participation with creating space for others to contribute. Doing so will further strengthen his effectiveness as a collaborator, mentor, and future research leader.

While the second half of the research work was one-to-one with the professor, the first half involved a small research cohort of 3-6 scholars. Describe the contribution that the scholar made to the research group discussion and learning:

Haorong consistently made insightful and impactful contributions to discussions, demonstrating exceptional intellectual curiosity and engagement at every opportunity. He was not only an active participant in conversations about his own research but also remained highly attentive to the work of his peers, regularly asking thoughtful, well-informed questions that reflected careful listening and deep consideration of the topics being presented. His contributions consistently showcased strong critical thinking, analytical depth, and the ability to evaluate ideas from multiple perspectives. Whether discussing technical concepts, research methodology, or broader implications, Haorong brought a level of engagement and intellectual maturity that elevated the quality of group discussions.

Would you recommend this scholar to a college admissions officer at a top college or university? Why or why not?

I give Haorong my strongest recommendation for admission to a highly selective university. Haorong is an extraordinary student whose intellectual ability, technical excellence, independence, and research maturity place him among the most impressive students I have worked with. Throughout the project, he consistently demonstrated the ability to master complex concepts, think critically about challenging problems, and produce high-quality work with minimal guidance. His combination of analytical depth, intellectual curiosity, and disciplined execution enabled him to make substantial contributions to a demanding research project that is typically explored at the upper-undergraduate or graduate level.

What distinguishes Haorong most is his rare ability to combine deep technical expertise with genuine curiosity and a relentless drive to learn. He engages thoughtfully with complex ideas, asks insightful questions, and consistently contributes meaningful perspectives that elevate discussions. Whether conducting independent research, evaluating technical tradeoffs, or engaging with the work of his peers, Haorong exhibits a level of maturity, initiative, and scholarly engagement that is uncommon among students at this stage. I am confident that he will thrive in a rigorous academic environment and make significant contributions to research, innovation, and the broader university community. Among the many talented students I have mentored, Haorong stands out as one with exceptional promise and the potential to excel at the highest levels of academia and research.

Evaluator's Signature:	
Evaluator's Printed Name:	Hanan Hibshi
Date of Evaluation:	September 30, 2026
Evaluator's Email:	hhibshi@andrew.cmu.edu

Pioneer Academics affirms that the scholar whose research paper is attached has followed Pioneer Academics' protocols for developing original research and that their paper has met Pioneer Academics' standards for authenticity. The grading and evaluation have met the standards collaboratively defined by Pioneer Academics and Oberlin College & Conservatory. If there are any questions regarding the work's academic integrity, please contact us at academic.integrity@pioneeracademics.com or by calling (855) 572-8863. We will conduct an immediate review and respond to your questions promptly.